KK-L-00159a(Army-QMC)

September 21, 1962
Interim Revision of
Fed. Spec. KK-L-159
December 6, 1956

INTERIM FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, (INSOLE)

This interim Federal specification was developed by the Quartermaster Research & Engineering Command, Natick, Massachusetts based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Service Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification KK-L-159, dated December 1956.

1. SCOPE

1.1 Scope.- This specification covers insoles made from cattlehide.

1.2 Classification.- The insoles shall be of the following types, tannages and classes as specified (see 6.2):

Type I - Combination tanned

Tannage a - Chrome - retanned

Tannage b - Vegetable - retanned

Type II - Straight vegetable

Class 1 - Men's

Class 2 - Women's

2. APPLICABLE SPECIFICATIONS, AND STANDARDS

2.1 Specifications and standards.- The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

O-L-164 - Leather Dressing: Mildew Preventive.

KK-L-311 - Leather, Methods of Sampling and Testing.

UU-P-268 - Paper, Kraft, Untreated, Wrapping.

FSC 8330

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PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid

PPP-B-591 - Boxes, Fiberboard, Wood-Cleated

PPP-B-601 - Boxes, Wood, Cleated Plywood

PPP-B-621 - Boxes, Wood, Nailed and Lock Corner

PPP-B-636 - Box, Fiberboard

PPP-T-60 - Tape, Pressure-sensitive Adhesive, Waterproof, For Packaging and Sealing.

FEDERAL STANDARDS:

FED. STD. NO. 102 - Preservation, Packaging and Packing Levels.

FED. STD. NO. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

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MILITARY SPECIFICATION

MIL-L-10547 - Liners, Case, Waterproof

MILITARY STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-

3.1.1 Type I.- The leather for the type I insole shall be cattlehide tanned by one of the following methods:

Tannage a - Tanned by a chrome tanning process followed by a vegetable retannage.

Tannage b - Tanned by a vegetable tannage process followed by a chrome retannage.

3.1.2 Type II.- The leather for the type II insole shall be cattlehide tanned with only vegetable tanning materials.

3.2 Treatment.- Unless otherwise specified (see 6.2) the leather for the type I and II insoles shall be treated with the fungicide paranitrophenol. The leather shall contain 0.18 to 0.60 percent of the fungicide, based upon the dry weight of the leather when tested as specified in 4.3.

3.3 Chemical requirements.-

3.3.1 Type I.- The leather for the type I insoles shall conform to the chemical requirements listed in table I, when tested as specified in 4.3.

TABLE I.- Chemical requirements

Characteristics	Minimum	Maximum
Degree of tannage 1/	35.0	-
Water soluble material percent 1/	-	26.0
Chloroform extract, percent 1/	1.8	10.0
Acidity, pH	3.0	4.0
ome, Cr_2O_3, percent 1/	2.0	-
Shrinkage temperature	90°C.	-

1/ Calculated to a moisture free basis.

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3.3.2 Type II.- The leather for the type II insole shall conform to the chemical requirements listed in table II, when tested as specified in 4.3.

TABLE II.- Chemical requirements

Characteristics	Minimum	Maximum
Degree of tannage 1/	60.0	-
Chloroform extract, percent 1/	3.0	10.0
Water soluble material, percent 1/	-	35.0
Insoluble ash, percent 1/	-	1.0
Acidity, pH	3.0	4.0

1/ Calculated to a moisture free basis.

3.4 Physical requirements.- The leather for type I and II insoles shall conform to the following physical requirements.-

3.4.1 Area stability.- The leather shall not shrink more than 10 percent in area when tested as specified in 4.3.

3.4.2 Cracking.- The leather shall not crack when tested as specified in 4.3.

3.4.3 Piping.- The leather shall not pipe when tested as specified in 4.3.

3.5 Thickness.- The thickness of the insoles shall be as specified (see 6.2). A plus tolerance of 1 iron shall be permitted and no minus tolerance shall be allowed.

3.6 Finish.- The insoles shall be finished full grain, degrained or buffed with the flesh side smooth and free from loose fibers.

3.7 Workmanship.- The insoles shall conform to the quality and grade of product established by this specification. The occurrrence of defects shall not exceed the applicable acceptable quality levels.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor inspection responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform y of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance.- Inspection shall be in accordance with Standard -STD-105, except where otherwise indicated herein.

4.2.1 Examination of the end item.- The classification of defects found during the examination shall be in accordance with the list shown in 4.2.1.1 and 4.2.1.2. The sample unit for these examinations shall be one insole. The lot size shall be expressed in terms of one insole. The inspection level and acceptable quality vel for these examinations shall be in accordance with 4.2.1.3.

4.2.1.1 Examination for visual characteristics.-

EXAMINE	DEFECTS
Brand	Brand on shank, or forepart
	Heavy brand in heel area
Wrinkles	Heavy fat wrinkles
Scratches	Deep scratches open or not healed
Grain damage	Grain damage which will not buff out
	Extensive cracked grain in shank or forepart
	Open tick scars not healed

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EXAMINE (cont'd)	DEFECTS
Grub holes	Open grub holes
	Healed grub scars in channel area.
Hardness	Hard and bony
Slaughter Cuts	Deep cuts
	Slight cuts parallel to and within the channel area
Flexibility	Excessive stiffness
	Flexibility not uniform
Pattern damage	Any pattern damage not filling the sole pattern.
Fiber characteristics	Hard and very tight
	Loose and spongy
Finish	Not full grain, degrained or buffed.
	Flesh side not smooth and free from loose fibers.

4.2.1.2 Examination for dimensional characteristics.-

EXAMINE	DEFECT
Thickness	Thickness of insoles not as specified. 1/

1/ The thickness shall be measured in accordance with method 1101 of Specification KK-L-311.

4.2.1.3 Inspection levels and Acceptable Quality Levels (AQL's).- The inspection levels and acceptable quality levels, expressed in defects per 100 units shall be as follows:

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EXAMINATION	INSPECTION LEVEL	AQL
For defects applicable to 4.2.1.1, visual characteristics	II	2.5
For defects applicable to 4.2.1.2, dimensional characteristics	L-6	4.0

4.2.2 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location sequence, or method of application.
Materials	Any component missing
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, inadequate stapling.
Content	Number of insoles per bundle is more or less than required. 1/
Weight	Net weight exceeds requirement.

1/ For this defect, two bundles from each shipping container in the sample shall be examined.

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4.3 End item testing.- The methods of testing specified in Specification KK-L-311, wherever applicable, and as listed in table III shall be followed except that pieces for physical and chemical tests shall be taken from each test unit (cut insole) in the sample for test, not closer than 1/2 inch to any edge. The sample unit shall be one (1) leather insole and the sample size shall consist of fifteen (15) leather insoles selected at random from the lot. When chemical requirements are specified on a moisture free basis, the method for moisture determination shall be Method 6221 of Specification KK-L-311. The lot shall be rejected if more than two (2) test failures occur for those requirements applicable to the sample unit or if any lot average fails to meet the specified requirement.

TABLE III.- Tests

Characteristics	Requirement Reference	Test Method	Requirement applicable to: Sample Unit	Lot Aver.
Material identification 1/	3.1.1 and 3.1.2	Std. Comm'l. 2/	X	-
Tannage 1/	3.1.1 and 3.1.2	Std. Comm'l. 2/	X	-
Paranitrophenol	3.2	4.3.1	-	X
Degree of tannage	3.3.1	6641 and 4.3.5	-	X
Water soluble material	3.3.1 and 3.3.2	6341	-	X
Chloroform, extract	3.3.1 and 3.3.2	6321	-	X
Acidity (pH)	3.3.1 and 3.3.2	6621	-	X
Chromic oxide	3.3.1	6511 or 4.3.2	-	X
Shrinkage temperature	3.3.1	4.3.3	-	X
Insoluble ash	3.3.2	6411	-	X
Area stability	3.4.1	4.3.4	X	-
Cracking	3.4.2	4021 3/	X	-
Piping	3.4.3	4111 3/	X	-

1/ The contractor shall submit a certificate of compliance for this characteristic.

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2/ One determination shall be made from each sample unit and the results shall be reported as "pass or fail".

3/ The diameter of the mandrel shall be 3 inches.

4.3.1 Test for the determination of percentage of paranitrophenol.- The percentage of paranitrophenol shall be determined in accordance with method 6711 of KK-L-311 except that two ml. of either chloroform extract shall be transferred to the ninth flask containing sodium citrate (20 ml. of a 10 percent solution of $Na_3 C_6 H_5 O_7$ in water). If the concentration of the developed paranitrophenol color is too intense for instrument readings, the 2 ml. of aliquot shall be suitably reduced. To compute the percent paranitrophenol on a dry basis use the following formula:

$$\text{Percent paranitrophenol (Dry basis)} = \frac{\text{Milligrams Paranitrophenol in aliquot X dilution}}{\text{10 X dry weight of specimen in grams}}$$

4.3.2 Chromic oxide - perchloric acid method.- The percentage of chromic oxide shall be determined in accordance with method 6515 of KK-L-311, except that a seventy two percent reagent grade perchloric acid shall be used in lieu of the 60 percent grade acid and a 40 percent solution phosphoric acid shall be used as follows: Just prior to the addition of 10 ml. of potassium iodide solution to the perchloric acid digest, add 30 ml. of the 40 percent phosphoric acid solution to complex the iron.

4.3.3 Shrinkage temperature.- The shrinkage temperature shall be determined in accordance with method 7011 of KK-L-311 except that the heating medium shall be water.

4.3.4 Area stability.- A 2 by 2 inch specimen shall be immersed in sufficient distilled water (68° to 78°F) to cover the specimen for 2 hours at room temperature (68° to 78°F) then remove blot on a towel and air dry at room temperature for 24 hours. The area shall then be measured and the change in area calculated

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to the nearest 0.1 percent. There shall be one determination per sample unit.

4.3.5 Degree of tannage.- The following applies only to type II leather; when a finishing treatment containing nitrogenous additives is used in the treatment of the vegetable-tanned type II leather, the supplier of insole leather shall submit for each lot a certification of the percent increase in nitrogen content (dry leather basis) resulting from the treatment. The certified percent increase in nitrogen shall be deducted from the total nitrogen found by analysis. The difference shall be used as the corrected nitrogen content for calculating degree of tannage.

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian agency procurements).

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Sixty pairs of insoles, of one type, tannage, class, and thickness only, shall be evenly stacked and then securely tied with cotton twine. When specified (see 6.2), ten insoles of one type, tannage, class, and thickness shall be completely wrapped in 30-pound minimum basis weight kraft paper conforming to grade B of Specification UU-P-268. The wrap shall be secured with cotton tape or twine, or pressure-sensitive tape.

5.1.2 Level C.- Insoles shall be packaged in accordance with the industry's practice.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A.- Insoles of one type, tannage, class, and thickness only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of Specification PPP-B-636. Each fiberboard shipping container shall have all seams, corners, and manufacturer's joint sealed by the application of 3-inch minimum width olive drab color cloth tape conforming to type III, class 1 of Specification PPP-T-60. Alternatively, shipping containers may be used that conform to style A or B, class 2 of Specification PP-B-576; overseas type of Specification PPP-B-591 or PPP-B-601; or class 2, style 2 or 4 of Specification PPP-B-621. Each alternative container shall be provided with a type I or II, grade B or C, class 2 sealed case liner conforming to Specification MIL-L-10547. Each shipping container shall be closed and reinforced with ...at steel strapping or tape banding in accordance with the appendix of the applicable container specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B.- Insoles of one type, tannage, class, and thickness only, ...ackaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of Specification PPP-B-636. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. Alternatively, shipping containers may be used that conform to style A or B, class 1 of Specification PPP-B-576; domestic type, style A or B of Specification ...P-B-591 or PPP-B-601; or class 1, style 2 or 4 of Specification PPP-B-621. The weight of the contents of each shipping container shall not exceed 65 pounds.

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5.2.3 Level C.- Insoles, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.-

5.3.1 Civilian agencies.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Standard No. 123.

5.3.2 Military requirements.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The leather insoles covered by this specification are intended for use in the types of footwear in which a leather insole is required.

6.2 Ordering data.- Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type, tannage and class required (see 1.2).

(c) When the fungicide paranitrophenol is not required (see 3.2).

(d) Thickness required (see 3.5).

(e) Whether ten insoles per package is desired (see 5.1.1).

(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 A suggested but not mandatory treatment of the finished leather insoles is as follows:

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a. Type I compound (hydrocarbon oil, cyclohexanone, perchlorethlene, and paranitrophenol described in Specification O-L-164.

b. When leather requiring paranitrophenol is specified, the treating of the leather may be conducted during the tanning and processing operation.

NOTICE.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procure-ment operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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